WearMoko

Flutter App

Installation & Setup Guide

Version	Flutter SDK
1.0.0	^3.5.3

Last Updated: June 07, 2026

For setup on Windows, macOS, and Linux

How to Use This Guide

This guide will walk you through everything you need to get the **WearMoko Flutter App** running — whether you're setting it up for the first time, sharing it with your team, or installing it on a new PC. You don't need to be an expert to follow along.

•	Follow the sections in order if this is your first time. If you just need to share the app or troubleshoot an issue, jump straight to the relevant section using the headings.

Section 1 — System Requirements

Before installing anything, make sure your computer meets these requirements.

Minimum Specs

Requirement	Minimum
Operating System	Windows 10+ | macOS 10.15+ | Linux
RAM	8 GB minimum (16 GB recommended)
Free Storage	At least 20 GB
Internet	Required — for Firebase, packages, and dependencies

Required Software

Install all of the following before proceeding. Each item links to its official download page.

1. Flutter SDK (v3.5.3 or higher)

This is the main framework the app is built on. Download it from: https://flutter.dev/docs/get-started/install

- Dart SDK is bundled with Flutter — no separate install needed.
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2. Git

Used to clone and manage the project code. Download: https://git-scm.com/

3. Code Editor (pick one)

- **Visual Studio Code** (recommended) — https://code.visualstudio.com/
  - Required VS Code extensions: Flutter, Dart, Firebase
- **Android Studio** — https://developer.android.com/studio

4. Android Development Tools

Required if you want to run or build for Android:

- Android Studio (install Android SDK inside it)
- Android SDK version 21 or higher
- JDK 11 or higher
- A physical Android phone (USB) OR a virtual device (emulator) created in Android Studio

5. iOS Development Tools (macOS only)

Only needed if you're on a Mac and want to build for iPhones/iPads.

- Xcode 12.0 or higher
- CocoaPods

6. Node.js (for Firebase backend functions)

Download the LTS version from: https://nodejs.org/

Section 2 — Initial Setup (First Time Only)

•	Only do this section once — when you're setting up the project for the first time on a new machine.

Step 1 — Verify Flutter is Installed

Open a terminal (Command Prompt on Windows, Terminal on Mac/Linux) and run:

```
flutter --version
dart --version
```

You should see version numbers printed. If you get an error saying command not found, Flutter is not in your PATH yet — see Troubleshooting at the end of this guide.

Step 2 — Get the Project Packages

Navigate into the project folder and download all dependencies:

```
cd wearmokoapp
flutter pub get
```

This may take a minute. Once it says 'Got dependencies!' you're good.

Step 3 — Set Up Firebase Configuration

The app uses Firebase as its backend. You have two options:

Create a New Firebase Project

1. Go to https://console.firebase.google.com/ and create a new project.
2. Add an Android app using package name: `com.example.wearmokoapp`
3. Download the `google-services.json` file and place it in `android/app/`
4. Repeat for iOS if needed, using the same bundle ID.

Step 4 — Configure Environment Variables

Open the .env file in the project's root folder and fill in your Firebase credentials:

```
FIREBASE_API_KEY=your_firebase_api_key
FIREBASE_AUTH_DOMAIN=your_project.firebaseapp.com
FIREBASE_PROJECT_ID=your_project_id
FIREBASE_STORAGE_BUCKET=your_bucket.appspot.com
FIREBASE_MESSAGING_SENDER_ID=your_sender_id
FIREBASE_APP_ID=your_app_id
```

•	Never share your .env file publicly. It contains private credentials. Add it to .gitignore if using GitHub.

Step 5 — Run Flutter Doctor

This command checks that everything is correctly installed and configured:

```
flutter doctor
```

Read through the output. A green checkmark ✓ means it's good. A red ✗ or yellow ! means something needs attention — fix those before proceeding.

Section 3 — Running the App Locally

Once setup is complete, pick one of the following methods to launch the app.

Option 1 — Android Emulator

Launch a virtual Android device first, then run the app:

```
flutter emulators --launch Pixel_4_API_30
flutter run
```

If you don't have an emulator set up yet, create one inside Android Studio under the AVD Manager.

Option 2 — Physical Android Phone

5. Enable Developer Mode on your phone: go to Settings > About Phone, then tap Build Number 7 times.
6. Enable USB Debugging in Settings > Developer Options.
7. Connect your phone to your PC with a USB cable.
8. Run the command below:

```
flutter run
```

Option 3 — iOS Simulator (macOS only)

```
open -a Simulator
flutter run
```

Option 4 — Physical iPhone / iPad (macOS only)

9. Connect your iOS device via USB.
10. Trust the developer certificate on the device when prompted.
11. Run:

```
flutter run
```

Option 5 — Debug Mode with Detailed Logs

Use this when something isn't working and you want to see what's happening under the hood:

```
flutter run -v
```

Section 4 — Building the App for Other Devices

Once you're ready to share or deploy the app, use these commands to generate installable files.

Building for Android

Standard APK (install directly on phones)

```
flutter build apk --release
```

Output file location: build/app/outputs/flutter-apk/app-release.apk

App Bundle (for Google Play Store uploads)

```
flutter build appbundle --release
```

Output file location: build/app/outputs/bundle/release/app-release.aab

Building for Windows

•	Requires Visual Studio 2022 with the 'Desktop development with C++' workload installed.

```
flutter build windows --release
```

Output location: build/windows/runner/Release/

Building for Web

```
flutter build web --release
```

Output location: build/web/

Section 5 — Installing on Another PC

There are several ways to share the WearMoko app with a teammate or install it on a new machine. Pick the method that best fits your situation.

Method 1 — Copy the Full Project (Recommended for Developers)

Best when the other person also wants to run or develop the app, not just use it.

12. Copy the entire project folder to the other PC.
13. Make sure Flutter SDK, Git, and Android Studio are installed on that PC.
14. Open a terminal in the project folder and run:

```
flutter pub get
flutter run
```

Method 2 — Share the APK File (Android Only)

Best for non-developers who just want to install and use the app on an Android phone.

15. On your PC, build the APK:

```
flutter build apk --release
```

16. Copy the `app-release.apk` file to the target Android device (via USB, Google Drive, email, etc.).
17. On the Android device, open the file in File Manager and tap Install.
18. If prompted, allow installation from unknown sources in your phone settings.

Method 3 — GitHub (Best for Teams)

Use this when multiple people are working on the project together.

First time — push to GitHub

```
git init
git add .
git commit -m "Initial commit"
git remote add origin https://github.com/yourusername/wearmokoapp.git
git branch -M main
git push -u origin main
```

On another PC — clone the repository

```
git clone https://github.com/yourusername/wearmokoapp.git
cd wearmokoapp
```

Then follow Section 2 (Initial Setup) on the new machine.

Method 4 — USB or Network Drive Transfer

19. Compress the project into a zip file:

```
# On Windows (PowerShell):
Compress-Archive -Path wearmokoapp -DestinationPath wearmokoapp.zip

# On macOS/Linux:
zip -r wearmokoapp.zip wearmokoapp
```

20. Transfer via USB drive, cloud storage, or network share.
21. Extract the zip and follow Section 2 (Initial Setup).

Section 6 — Firebase Configuration

Setting Up Firebase Emulator (Optional)

The Firebase Emulator lets you test the app locally without connecting to your live Firebase project. Useful during development.

Step 1 — Install Firebase CLI

```
npm install -g firebase-tools
```

Step 2 — Log In

```
firebase login
```

Step 3 — Start the Emulator

```
firebase emulators:start
```

•	Remember to update your Firebase initialization in main.dart to point to the emulator host while testing, then revert before building for production.

Deploying Firebase Functions

```
cd functions
npm install
firebase deploy --only functions
```

Section 7 — Troubleshooting

Something not working? Find your error below and follow the fix.

Error / Symptom	Likely Cause	How to Fix
"Flutter command not found"	Flutter is not in your system PATH.	Windows: Add C:\flutter\bin to your environment variables. Mac/Linux: Open your terminal config file (~/.bashrc or ~/.zshrc) and add: `export PATH="$PATH:~/flutter/bin"` then run: source ~/.bashrc
"Android SDK not found"	Flutter can't locate the Android SDK on your machine.	`flutter config --android-sdk /path/to/android-sdk` (Windows path is usually: C:\Users\YourName\AppData\Local\Android\sdk)

Gradle build fails	Corrupted cache or outdated build files.	`flutter clean`
		`flutter pub get`
		`flutter build apk`

"Unable to find google-services.json"	The Firebase config file is missing or in the wrong location.	Place google-services.json in the android/app/ folder, then run flutter clean and try again.
Firebase initialization fails	Incorrect or missing credentials.	Check that your .env file has the correct Firebase values. Confirm google-services.json is in android/app/. Verify the Firebase project is active in the Firebase Console.

App crashes on startup	An error is occurring at launch — need more details.	`flutter run -v`
		
		For Android, check device logs:
		`adb logcat`

"Location permission denied"	The app's permissions were not granted on the device.	Go to Settings > Apps > WearMoko > Permissions and enable Location, Camera, and Contacts.
Cannot connect to Firebase	Network or credentials issue.	Check your internet connection. Verify Firebase credentials in .env. Check Firebase security rules in the Firebase Console.
iOS build fails (macOS only)	CocoaPods or Xcode	`cd ios` `pod repo update`

	configuration issue.	pod install cd .. flutter clean flutter pub get flutter run

Section 8 — App Permissions

The WearMoko app will ask for the following permissions when you first open it on Android. These are required for full functionality.

Permission	Why It's Needed
Location	Fine and coarse location access for device/asset tracking.
Camera	Capturing profile pictures and in-app photos.
Contacts	Displaying and sharing contact information within the app.
Notifications	Push notifications and real-time alerts.
Storage	Accessing files, images, and videos saved on the device.

Section 9 — Pre-Run Checklist

Use this checklist before running the app on a new PC to make sure nothing is missed.

	Flutter SDK installed and added to PATH
	Git installed
	Android SDK and Android Studio installed (for Android builds)
	Xcode installed (for iOS builds on macOS)
	.env file configured with correct Firebase credentials
	google-services.json placed in android/app/
	flutter pub get completed successfully
	flutter doctor shows no errors (all green checkmarks)
	Internet connection available
	At least 20 GB of free disk space confirmed

Additional Resources

If you need more help, these official resources are the best place to look:

- **Flutter Docs:** https://flutter.dev/docs
- **Firebase Flutter Setup:** https://firebase.flutter.dev/
- **Android Development:** https://developer.android.com/
- **iOS Development:** https://developer.apple.com/ios/
- **Flutter Issues on GitHub:** https://github.com/flutter/flutter/issues
- **Stack Overflow (Flutter tag):** https://stackoverflow.com/questions/tagged/flutter